

# EC-350 AI and Decision Support Systems

---

## Week 3 Solving Problems by Searching

Dr. Arslan Shaukat



Acknowledgement: Lecture slides material from  
Stuart Russell

1

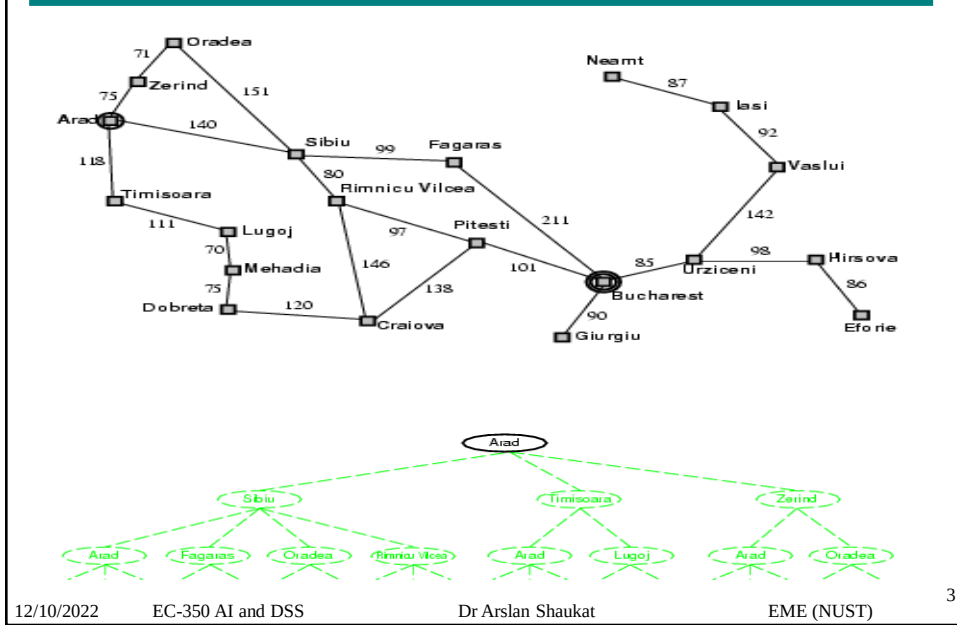
## Searching for Solutions

---

- We solve problems by searching the state space
- State space is represented by a graph or a tree
- We start at root node, check if it is a goal state
- If not, apply the Result/successor function to generate new state (node)
- The choice of which state to expand is determined by the search strategy

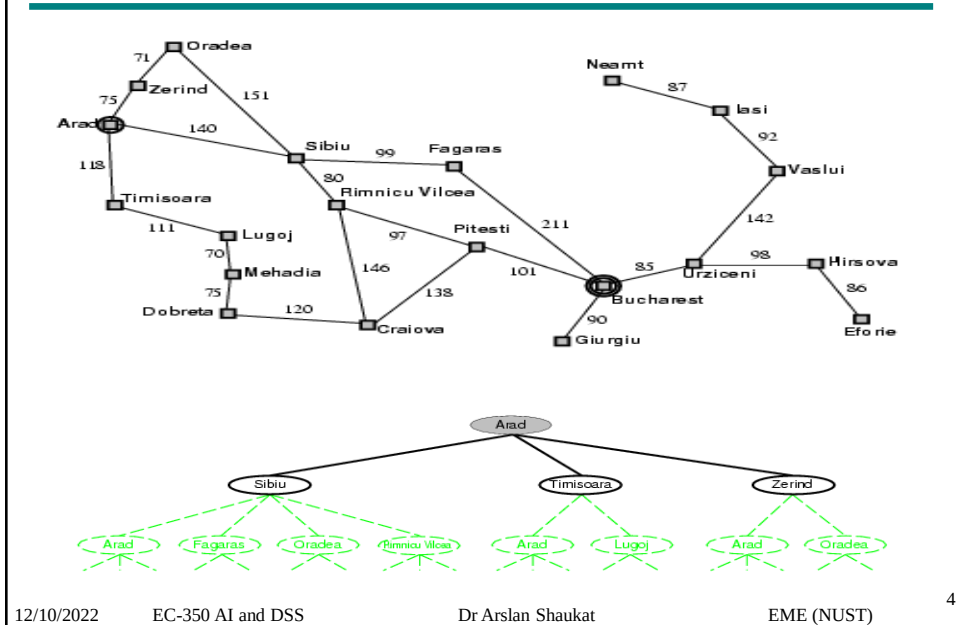
2

## Tree Search Example



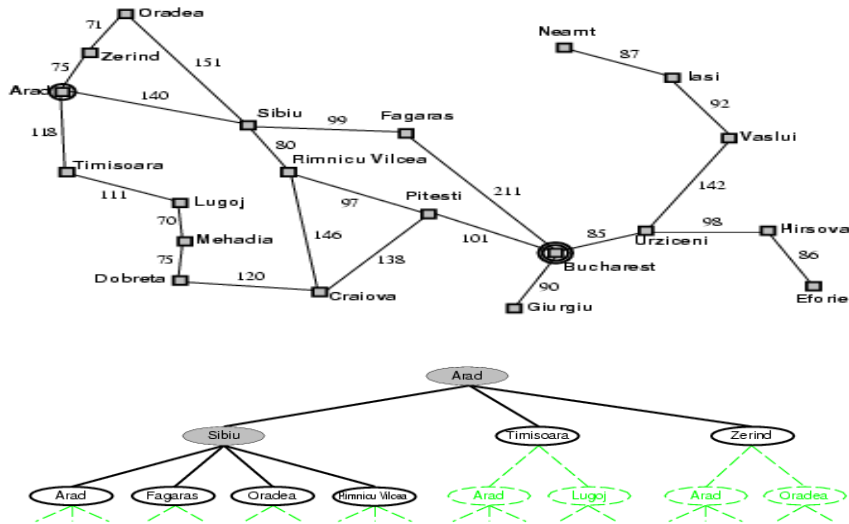
3

## Tree Search Example



4

## Tree Search Example



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

EME (NUST)

5

5

## Assumptions about Node

- Node is a data structure with five components
  - *State*: the state in the state space to which node correspond
  - *Parent node*: the node which generated this node
  - *Action*: the action that was applied to parent to generate this
  - *Path-cost*: the cost, denoted by  $g(n)$ , from initial state to this node
  - *Depth*: the number of steps along the path from initial state to the goal

12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

EME (NUST)

6

6

## Fringe

---

- We shall use a data structure called “fringe”
- This will contain the collection of nodes which has been generated but not expanded
- The search strategy would decide next node to be expanded from this list
- Assume collection of nodes is represented as a Queue, Stack etc.

## Search Strategies Evaluation

---

- Strategies are evaluated along the following dimensions:
  - *completeness: does it always find a solution if one exists?*
  - *time complexity: how long does it take*
  - *space complexity: how much memory is needed*
  - *optimality: does it always find a least-cost solution?*
- Time and space complexity are measured in terms of
  - *b: maximum branching factor of the search tree*
  - *d: depth of the shallowest goal-node*
  - *m: maximum depth of the tree (may be  $\infty$ )*

## Uninformed Search Strategies

- Uninformed (Blind) search strategies use only the information available in the problem definition
  - *Breadth-first Search*
  - *Uniform-cost search*
  - *Depth-first search*
  - *Depth-limited search*
  - *Iterative deepening search*

12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

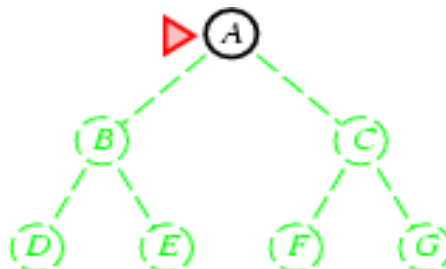
EME (NUST)

9

9

## Breadth-first Search

- Expand shallowest unexpanded node first
- **Implementation:**
  - Put the newly expanded node at the back of the fringe
  - use the fringe as a *Queue*, FIFO
  - Fringe = A



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

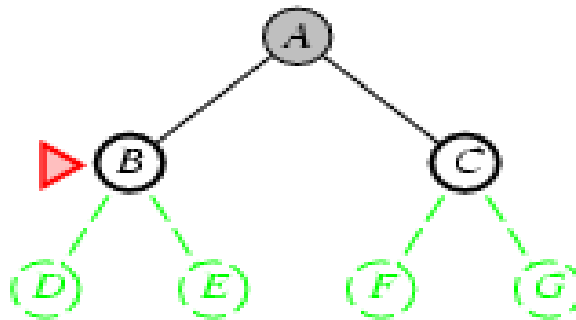
EME (NUST)

10

10

## Breadth-first Search

- Expand shallowest unexpanded node first
- Check if A is goal, it is not, expand A
- Fringe = B,C



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

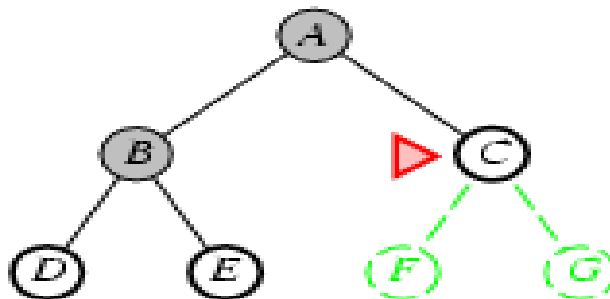
EME (NUST)

11

11

## Breadth-first Search

- Check if B is goal, it is not, expand B
- Fringe = C, D, E
- Check if C is goal, it is not, expand C
- Fringe = D, E, F, G



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

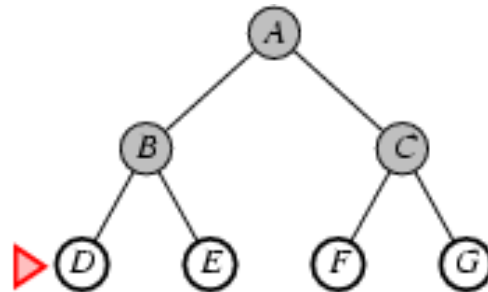
EME (NUST)

12

12

## Breadth-first Search

- Expand shallowest unexpanded node
- Check if D is goal, it is not, no children
- Fringe = E, F, G
- Search goes on until goal is found. Once it is found, the search terminates



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

EME (NUST)

13

13

## Properties of Breadth-first Search

- **Complete:** Yes (if  $b$  is finite)
- **Time:**  $1 + b + b^2 + b^3 + \dots + b^d = O(b^d)$
- **Space:**  $O(b^d)$  (keeps every node in memory)
- **Optimal:** Yes (if cost = 1 per step)
- **Space** is the bigger problem (more than time)

12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

EME (NUST)

14

14

## Breadth-first Search

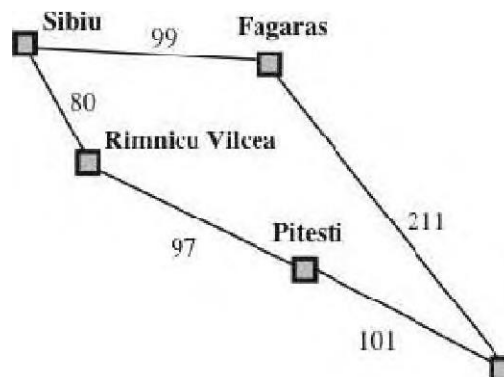
| Depth | Nodes     | Time      | Memory  |
|-------|-----------|-----------|---------|
| 2     | 111       | .11 mSecs | 107 KB  |
| 4     | 11,111    | 11 mSecs  | 10.6 MB |
| 6     | $10^6$    | 1.1 Secs  | 1 GB    |
| 8     | $10^8$    | 2 Mins    | 103 GB  |
| 10    | $10^{10}$ | 3 Hours   | 10 TB   |
| 12    | $10^{12}$ | 13 Days   | 1 PB    |
| 14    | $10^{14}$ | 3.5 Years | 1 XB    |
| 16    | $10^{16}$ | 350 Years | 10 XB   |

Time & Memory requirements for BFS. The numbers shown assume branching factor  $b = 10$ ; 1 million nodes per second expansion; 1,000 bytes per node

15

## Uniform-cost Search

- Expands a node with the lowest path cost
- Equivalent to breadth-first if all step costs are equal
- Implementation:
  - *fringe = queue ordered by path cost*



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

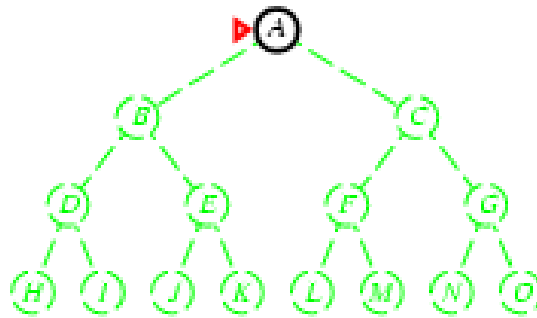
EME (NUST)

16

16

## Depth-first Search

- Expand deepest unexpanded node
- **Implementation:**
  - Put the expanded node at the *front* of the fringe
  - use fringe as a *Stack*, LIFO
  - Fringe = A



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

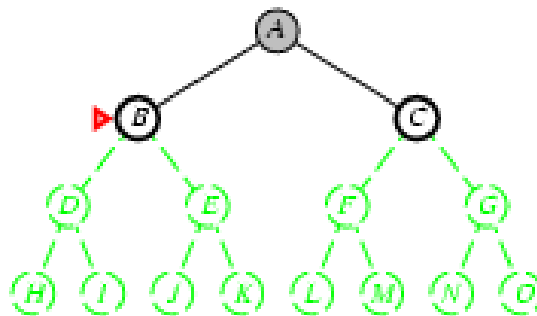
EME (NUST)

17

17

## Depth-first Search

- Expand deepest unexpanded node
- Check if A is goal, it is not, expand A
- Fringe = B,C



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

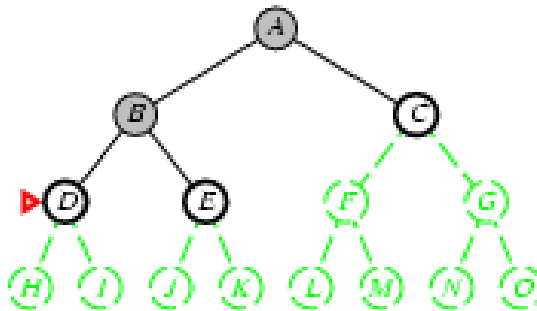
EME (NUST)

18

18

## Depth-first Search

- Check if B is goal, it is not, expand B
- Fringe = D,E,C



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

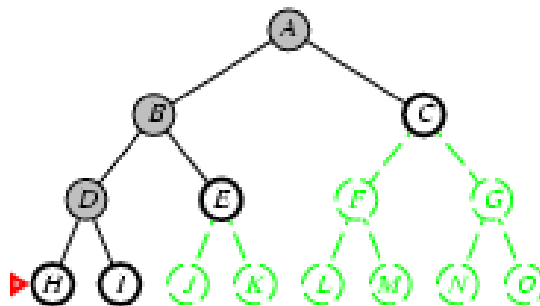
EME (NUST)

19

19

## Depth-first Search

- Check if D is goal, it is not, expand D
- Fringe = H, I, E, C



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

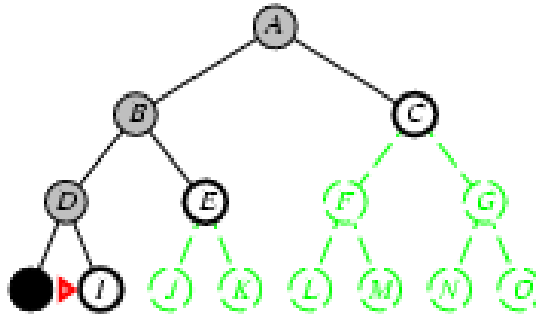
EME (NUST)

20

20

## Depth-first Search

- Both H and I has not successors so remove them if it is not goal
- Fringe = E, C



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

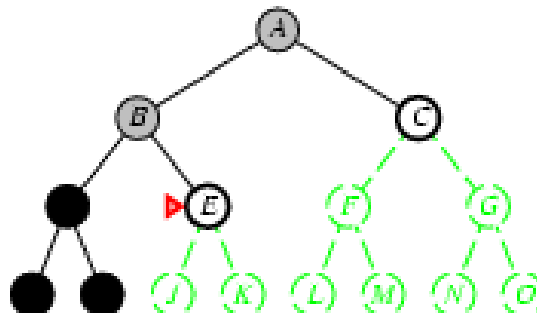
EME (NUST)

21

21

## Depth-first Search

- Check if E is goal, it is not, expand E
- Fringe = J, K, C



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

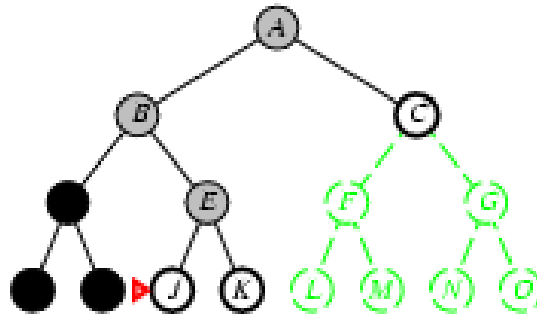
EME (NUST)

22

22

## Depth-first Search

- J and K do not have successors neither is any of them goal so remove them from fringe one by one
- Fringe = C



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

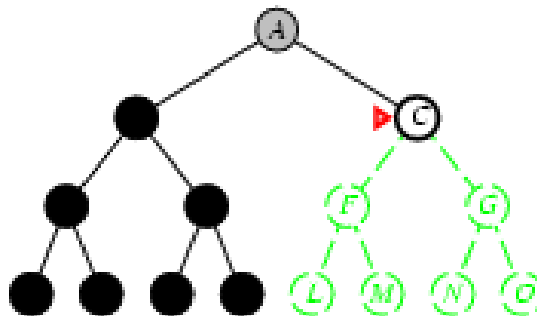
EME (NUST)

23

23

## Depth-first Search

- Check if C is goal, it is not, expand C
- Fringe = F, G



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

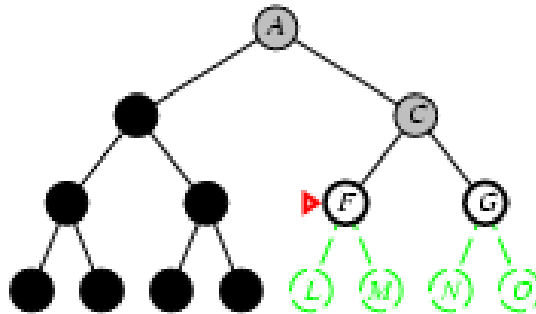
EME (NUST)

24

24

## Depth-first Search

- Check if F is goal, it is not, expand F
- Fringe = L, M, G



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

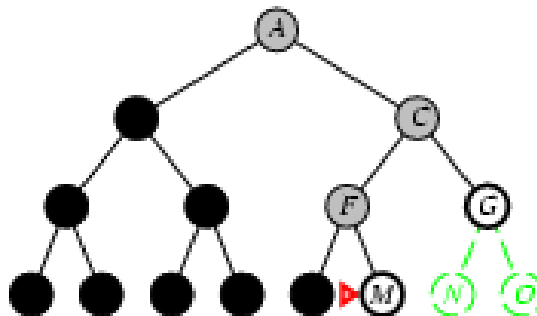
EME (NUST)

25

25

## Depth-first Search

- Both L, M are not goal, neither they have any successor
- Remove L, M from fringe
- Fringe = G
- Check if G is goal, it is
- The search terminates



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

EME (NUST)

26

26

## Properties of Depth-first Search

- Complete? No: fails in infinite-depth spaces, spaces with loops
  - *Complete in Finite state space*
  - *Modify to avoid repeated states along path*
- Time?  $O(b^m)$ : terrible if  $m$  is much larger than  $d$
- Space?  $O(bm)$
- Optimal? No

12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

EME (NUST)

27

27

## Depth-Limited Search

- Depth-first search with depth limit  $l$ , i.e., nodes at depth  $l$  have no successors
- Introduces an additional source of incompleteness if we choose  $l < d$
- DFS can be viewed as a special case of depth-limited search with  $l = \infty$ .

12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

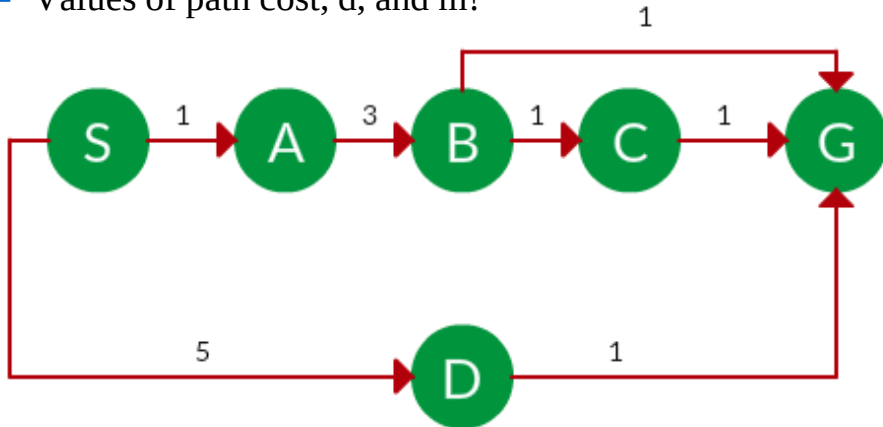
EME (NUST)

28

28

## Example

- Find the path from S to G
- Use BFS, UCS and DFS
- Values of path cost, d, and m?



12/10/2022

EC-350 AI and DSS

Dr Arslan Shaukat

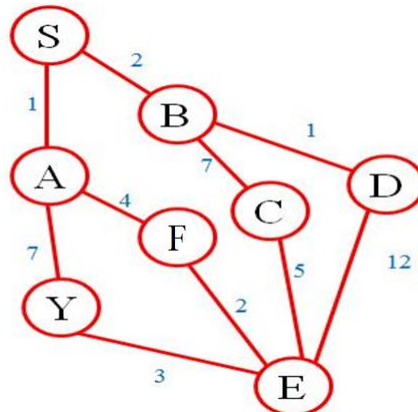
EME (NUST)

29

29

## Example

- Find path from S to E
- Use
  - BFS
  - UCS
  - DFS
- Values of Path cost, d, and m?



12/10/2022

EC-342 AI and DSS

Dr Arslan Shaukat

EME (NUST)

30

30

## Assignment # 1

---

- Deliverables:
  - *Part I to be solved on paper*
  - *Print out of Code in Part II with Comments and Output*
- Due after 2 weeks